



Vectors - Exercises Solutions

Exercise 8.1. In the statement below, underline x whenever it refers to the vector.

"The vector x is defined as $x = (x_1, x_2, x_3)$. The x -coordinate of x is x_1 ."

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Exercise 8.2. Calculate the magnitudes of the following vectors:

(a) $\mathbf{p} = (3, 4)$

$$\|\mathbf{p}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

(b) $\mathbf{u} = (1, 2, 3)$

$$\|\mathbf{u}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14} =$$

(c) $\mathbf{v} = (5, -12, 0)$

$$\|\mathbf{v}\| = \sqrt{5^2 + (-12)^2 + 0^2} = \sqrt{25 + 144 + 0} = \sqrt{169} = 13$$

(d) $\mathbf{w} = (1, 0, 1)$

$$\|\mathbf{w}\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{1 + 0 + 1} = \sqrt{2}$$

Exercise 8.3. Let $\mathbf{u} = (2, 3)$, $\mathbf{v} = (3, -2)$ and $\mathbf{w} = (1, 6)$. Write down

(a) $\mathbf{u} + \mathbf{v}$

$$\mathbf{u} + \mathbf{v} = (2, 3) + (3, -2) = (2 + 3, 3 + (-2)) = (5, 1)$$

(b) $\mathbf{u} + \mathbf{w}$

$$\mathbf{u} + \mathbf{w} = (2, 3) + (1, 6) = (2 + 1, 3 + 6) = (3, 9)$$

(c) $\mathbf{v} - \mathbf{w}$

$$\mathbf{v} - \mathbf{w} = (3, -2) - (1, 6) = (3 - 1, (-2) - 6) = (2, -8)$$

Exercise 8.4. Calculate $k\mathbf{v}$, where $\mathbf{v} = (2, 3, 4)$ and

(a) $k = 2$

$$k\mathbf{v} = 2 \times (2, 3, 4) = (4, 6, 8)$$

(b) $k = 1.5$

$$k\mathbf{v} = 1.5 \times (2, 3, 4) = (3, 4.5, 6)$$

(c) $k = 1$

$$k\mathbf{v} = 1 \times (2, 3, 4) = (2, 3, 4)$$

(d) $k = 0$

$$k\mathbf{v} = 0 \times (2, 3, 4) = (0, 0, 0)$$

Exercise 8.5. Which pairs of the following vectors are perpendicular?

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \quad \mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

$$\mathbf{u} \cdot \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} = -1 + 4 - 3 = 0$$

$$\mathbf{u} \cdot \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = 2 - 6 + 3 = -1$$

$$\mathbf{v} \cdot \mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = -2 - 6 - 1 = -9$$

Therefore $\mathbf{u} \perp \mathbf{v}$.